

Automated standardization and harmonization of laboratory units in large-scale clinical data using open-source R functions

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ARTICLE INFO

Keywords:

Electronic medical records
Clinical laboratory data
Data quality
Algorithms
Data preprocessing

ABSTRACT

Background and Objective: Clinical laboratory data constitute a major component of electronic medical records, yet their secondary use is hindered by inconsistent unit representations, especially in heterogeneous multi-source databases. While UCUM and LOINC standards exist, real-world datasets often contain non-standard unit strings and fragmented mappings, limiting interoperability and introducing analytic bias. This study aimed to develop and evaluate open-source functions for automated standardization of units to UCUM-valid formats and harmonization of laboratory results to reference units within LOINC groups.

Methods: We developed and evaluated two scalable open-source R functions: one for standardizing heterogeneous unit expressions into valid UCUM codes using preprocessing, deterministic mapping, and rule-based correction; and another for harmonizing UCUM-standardized units and values to designated SI or Conventional reference units. Both functions were applied to 163.9 million LOINC-mapped quantitative results from the Intego-II primary care database. Validation involved UCUM web service checks.

Results: Among 163,946,741 records with 2,019 distinct unit strings, 157,874,970 (96.2%) were standardized, reducing unit heterogeneity by 81% (from 2,019 to 381 UCUM-valid units). Before applying the function, only 70% of records included UCUM-valid units. Validation confirmed 99.97% correctness of standardized units. Harmonization converted 83.4% of standardized records to reference units within two minutes, consolidating them into 39–42 distinct units per system. Approximately 33% required numeric conversion. All conversions matched UCUM API outputs with 100% concordance rate. Unharmonized cases (16.6%) were primarily due to dimensional mismatches or incompatible arbitrary units. Harmonization also corrected LOINC property inconsistencies in 2.8 million records and reduced code fragmentation, improving distributional characteristics in up to 23% of LOINC groups.

Conclusions: Automated, scalable standardization and harmonization of laboratory units are feasible and important, substantially improving consistency and interoperability of large multi-source datasets. These functions enable reliable data integration for analytics and machine learning, reduce bias from coding and unit variability, and provide a quality-control mechanism for identifying structural inconsistencies. The open-source implementation supports integration into routine ETL pipelines and advances the reuse of real-world laboratory data.

1. Introduction

Clinical laboratory data account for approximately 70 % of the information contained in Electronic Medical Records (EMRs), with the majority comprising quantitative results [1]. Each quantitative result is more than just a number—it represents the product of a measurement value and its associated unit. Without standardized and clearly defined

units, result values can become ambiguous or even misleading [2].

While individual lab reports are typically interpretable within their local context, retrospective and multi-source laboratory datasets pose significant challenges. Data collected from heterogeneous sources often lack consistent unit formats and coding structures, making them difficult to integrate, compare, or analyze reliably [3]. This heterogeneity directly impacts secondary data uses, such as population-level research,

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clinical decision support, and Machine Learning (ML) applications.

To address this, widely adopted standards have been developed: the Logical Observation Identifiers Names and Codes (LOINC) system for uniquely identifying lab tests [4], and the Unified Code for Units of Measure (UCUM) for representing measurement units in a consistent, machine-readable form. UCUM, in particular ensures clear, unambiguous electronic communication of quantities and their units. It enables automated validation, parsing, and conversion of units, making it a cornerstone for scalable data exchange and computational analysis [5]. Recognizing its importance, both European and United States regulatory and technical standards now require or recommend UCUM as the preferred unit system for clinical laboratory data exchange [6,7].

However, a critical limitation remains: UCUM and LOINC are not intrinsically linked. A single LOINC code can be used with multiple units in Real-World Data (RWD)—including free-text or improperly formatted units that are invalid under UCUM syntax [4,8]. Meanwhile, UCUM allows a wide range of valid expressions for the same unit, which adds to the variability. This flexibility may be tolerable in clinical reporting, but it becomes a serious obstacle for downstream data processing, analytics, and interoperability. Adding to the complexity, laboratory tests that are clinically or analytically similar are often represented by distinct, highly specific (granular) LOINC codes, fragmenting conceptually unified features.

To introduce a concrete use case, these challenges represent sources of embedded bias introduced into clinical ML models [9]. Feature engineering—a crucial step in ML model development—is particularly sensitive to inconsistencies in test codes and units. Fragmentation of clinically equivalent tests across multiple LOINC codes leads to incomplete or overlapping feature definitions, while inconsistent and non-standard units introduce errors in numeric values, potentially distorting learned associations or thresholds [10]. Together, these issues reduce model generalizability, compromise reproducibility, and limit the scalability of ML models across institutions and real-world clinical deployments.

In the context of big data analytics and clinical ML pipelines, standardizing and harmonizing related lab codes and units is essential—but currently unsupported by standard tools. This work targets two key gaps that hinder the effective secondary use of quantitative lab data. First, while the National Library of Medicine (NLM) provides resources validating whether a unit is a valid UCUM unit expression, there is no existing tool to standardize invalid unit expressions into valid UCUM format—a basic requirement for automation and machine-readability [11,12]. Second, we aim to establish reference unit mappings not just at the individual LOINC code level, but at the group level, linking clusters of related LOINC codes to harmonized reference units.

To address these gaps, we introduce two novel open-source R functions that we developed de novo as part of the *lab2clean* package: one for standardizing real-world units into valid UCUM expressions, and another for harmonizing result values and units across LOINC groups by converting between UCUM-compatible units. We apply and test these tools on a large, heterogeneous real-world database comprising over 160 million quantitative laboratory records from the Intego-II primary care database, evaluating their practical relevance in improving the consistency, comparability, and analytic utility of heterogeneous retrospective laboratory data. The presented R functions were added to version 2 of the open-source R-package “lab2clean”, available on CRAN at <https://cran.r-project.org/package=lab2clean>.

2. Methods

2.1. Standardization function – ensuring UCUM validity

To address the challenge of non-standard and syntactically invalid unit expressions commonly found in real-world laboratory datasets, we developed this function to transform heterogeneous unit representations into valid, UCUM-compliant formats (Fig. 1).

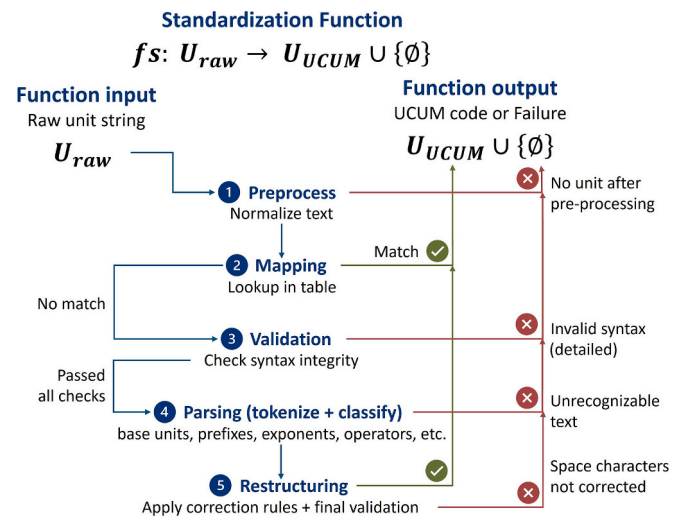


Fig. 1. Standardization function workflow. The standardization function $fs: U_{raw} \rightarrow U_{UCUM} \cup \{\emptyset\}$ maps raw unit strings (U_{raw}) to valid UCUM expressions (U_{UCUM}) or returns a failure case ($\emptyset$). The workflow involves five sequential steps: (1) preprocessing (normalization of text), (2) mapping to a curated synonym table, (3) validation of syntax integrity, (4) parsing of tokens (base units, prefixes, exponents, operators, etc.), and (5) restructuring with correction rules and final validation. Successful matches return UCUM codes, while failures are triggered by invalid syntax, unrecognizable text, or unresolved errors such as space characters.

The standardization workflow begins with extensive preprocessing, where raw unit strings are cleaned and normalized. Following preprocessing, cleaned unit strings are matched against an internally curated mapping table, which consists of 3,727 real-world non-standard synonyms linked to 1,455 common UCUM-standardized expressions. This mapping table is curated from different available resources/lists [4,13–15].

For unit expressions not captured in the mapping table, the function proceeds to a validation phase. Here, a series of syntactic integrity checks leveraging R’s base regular expression functions are applied to flag malformed patterns. Each detected anomaly is annotated in the `cleaning_comments` column, aiding transparency and potential manual review.

Unit expressions that are not clearly invalid but remain unmapped proceed to a parsing and semantic reconstruction phase. The unit string is tokenized based on separators such as operators and parentheses. Tokens are evaluated and reclassified into semantic groups (e.g., UCUM base units, numeric exponents, annotations, SI prefixes), and a series of correction rules are then iteratively applied. After correction, the restructured unit string undergoes a second round of validation. Expressions passing this final check are assigned as standardized UCUM codes.

2.2. Harmonization function – linking reference UCUM units with LOINC groups

To ensure the comparability and aggregability of quantitative laboratory results across heterogeneous data sources, we developed a second function, `harmonize_lab_unit`. This function harmonizes both unit expressions and associated numeric result values to a designated reference unit for each LOINC-defined test or group (Fig. 2) [13].

The conversion logic of the function relies on the accurate determination of each unit’s dimension and magnitude, as defined by UCUM. To this end, a supporting function extracts these parameters by parsing the UCUM unit expression into a vector representation that encodes the unit’s dimension as combinations of base units raised to certain powers as well as its magnitude or numeric scaling factor.

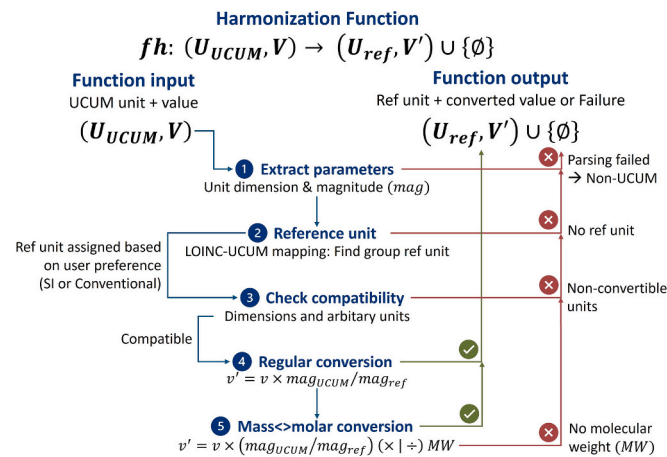


Fig. 2. Harmonization function workflow. The harmonization function $fh: (U_{UCUM}, V) \rightarrow (U_{ref}, V') \cup \{\emptyset\}$ maps standardized UCUM unit-value pairs to reference unit-value pairs or failure $\{\emptyset\}$. Input units (U_{UCUM}) are processed for (1) extracting unit parameters (dimension and magnitude), (2) identifying the designated reference unit (U_{ref}) from a LOINC-UCUM mapping table, and (3) checking compatibility (dimensions and arbitrary units). If compatible, original values (V) are transformed to the converted value (V') either (4) regular conversion or (5) mass<>molar conversion using analyte-specific molecular weight (MW). Failure occurs if the unit is non-UCUM, no reference unit is available, units are incompatible, or molecular weight is missing.

Users may specify a preferred harmonization system: the International System of Units (SI) or the Conventional system. The function then standardizes the units accordingly, performing numeric conversions when required. In cases involving a change in the underlying measurement property (e.g., from mass concentration to substance concentration), the LOINC code is also updated to reflect this transformation.

Harmonization is guided by a curated LOINC-UCUM mapping table, which defines the reference unit for each LOINC group in both SI and Conventional systems. This table was constructed by compiling all example units listed in the official LOINC release—the only formal linkage between LOINC codes and UCUM units. These examples were filtered and clustered by dimension and magnitude to identify groups of equivalent units. Within each group, a single reference unit was selected based on one or both of the following criteria: (1) adherence to measurement standards recommended by the IFCC-IUPAC Committee on Nomenclature, Properties, and Units (C-NPU) [14], and (2) linkage to a valid OMOP concept ID, ensuring compatibility with standardized health data models.

Once the appropriate reference unit is identified for a given test, conversion logic is then determined based on the dimension and magnitude attributes of the source and target units. If the conversion need is confirmed, two conversion cases are handled: (1) Regular unit conversion: when units are dimensionally equivalent and differ only in magnitude (e.g., mg/dL to g/L). (2) Mass-molar conversion: for analytes with known molecular weights, which are frequent in electrolyte and enzymatic assays. Necessary molecular weight data of laboratory analytes were collected programmatically by APIs from the PubChem database by the National Institutes of Health (NIH) [15]. To this end, units are flagged as non-convertible if they are incompatible (e.g., due to mismatched dimensions), involve arbitrary units that differ in symbolic reference (e.g., [IU] vs. [arb.U]).

2.3. Evaluation database: Intego-II primary care database

We evaluated the developed functions on the laboratory data table of the Intego-II primary care database, a large-scale Belgian, multi-centric database built upon more than three decades of pseudonymized

electronic medical record (EMR) data collection from general practices in Flanders, Belgium [16]. For this study, we accessed the December 2024 release, which included contributions from 180 participating practices, covering approximately one million patients [17]. Intego-II aggregates laboratory results originating from multiple external laboratories serving these practices, introducing substantial heterogeneity in unit representations and coding practices.

For evaluating how harmonization affected the distribution of laboratory results, we applied three evaluation metrics, each designed to capture a different aspect of distributional quality: (1) Interquartile Range (IQR), representing the width of the middle 50 % of observations, (2) Outlier proportion, defined as the percentage of observations falling outside the 5th–95th percentile range, and (3) Mode count, calculated by kernel density smoothing to estimate the number of peaks in the distribution. These measures capture whether harmonization reduces spurious variability and fragmentation caused by inconsistent unit scaling or incorrect mappings. In practice, a narrower IQR after harmonization indicates reduced variability in values for the same analyte, suggesting improved comparability across sources. A lower outlier proportion signals fewer spurious or implausible values, reflecting improved quality. For mode count, a reduction toward unimodality indicates successful consolidation of fragmented unit scales, whereas an increase may suggest introduced inconsistencies. Although there is no single ‘ideal’ value across all analytes, improvement is characterized by smaller IQR, fewer outliers, and reduced multimodality.

3. Results

The December 2024 release of the Intego-II primary care database contained approximately 223 million laboratory test records. Among these, 180 million results ($\approx 81\%$) were identified as quantitative measurements, highlighting the dominant role of numerical data in clinical laboratory data workflows. Of the quantitative results, approximately 164 million records ($\approx 91\%$) were successfully mapped to LOINC codes, providing the necessary linkage to enable subsequent harmonization processes.

All processing was performed in R version 4.4.1 on a virtual machine with 8 processing cores, and 128 GB of RAM, running Windows Server Datacenter. Standardization of 163.9 million records completed in under 10 s; harmonization in under 2 min. The Algorithmic complexity is approximately $O(d_n)$, where d_n denotes the number of distinct unit strings (for standardization) or distinct unit-value combinations (for harmonization). Since d_n is typically several orders of magnitude smaller than the total number of records, runtime is driven by the diversity of unit expressions rather than dataset size. Subsequent application of mappings across all records is vectorized, keeping runtime effectively constant with respect to dataset size. This design ensures scalability to national-scale laboratory datasets with hundreds of millions of records.

3.1. Performance of the unit standardization function

Among the 163,946,741 LOINC-mapped quantitative results, a total of 2,019 distinct unit expressions were identified (Table 1). After applying the standardization function, 157,874,970 records (96.2 %) were successfully standardized, with 830 real-world units transformed into 381 unique valid UCUM unit expressions in less than 10 s. This represents a reduction in unit heterogeneity of approximately 81 % (from 2,019 to 382).

Notably, 115,150,773 records (70.2 %) already contained UCUM-valid units before applying the function. The function therefore improved an additional 26.0 % of records, which were converted into UCUM-compliant formats through its preprocessing, mapping, and rule-based correction steps. A total of 6,228,779 records (3.8 %), corresponding to 1,194 unique unit expressions (59.1 %), could not be standardized due to severe formatting or semantic inconsistencies.

Importantly, to assess the correctness of the function’s output, we

Table 1

Results of the Unit Standardization Function Applied to Intego-II Laboratory Data.

	Result records		Distinct source units	
Total results	163,946,741	(%)	2,019	(%)
Total standardized	157,692,773	96.2 %	824	40.8 %
UCUM-valid source unit	115,150,773	70.2 %	291	14.4 %
Standardized to UCUM	42,541,915	26.0 %	533	26.4 %
Total not standardized	6,254,053	3.8 %	1,195	59.2 %
unrecognizable text	5,721,848	3.5 %	434	21.5 %
unrecognizable text, space characters	241,605	0.1 %	434	21.5 %
multiple operators in a row (./)	137,841	0.1 %	26	1.3 %
space characters	82,467	0.1 %	40	2.0 %
+/- operations	46,603	0.0 %	238	11.8 %
No unit	16,137	0.0 %	10	0.5 %
unbalanced parentheses	7,549	0.0 %	12	0.6 %
digits before parenthesis	3	0.0 %	1	0.0 %

validated all UCUM units generated by the function using the official UCUM web service. Among the unit records standardized by the function to UCUM-valid expressions, 99.97 % were confirmed as truly valid UCUM expressions, corresponding to 94.3 % of the standardized distinct units.

These findings demonstrate that while the majority of records were initially UCUM-compliant, a substantial proportion—over 42 million records—benefited from automated correction, significantly improving the consistency and machine-readability of unit expressions across this large-scale, multi-source dataset.

Summary of unit standardization outcomes across 163.9 million quantitative laboratory result records and 2,019 distinct source unit strings. The majority (96.2 %) of records were successfully standardized to UCUM-compliant formats, either because the source unit was already UCUM-valid (70.2 %) or required transformation (26.4 %). Unstandardized cases (3.8 % of records) primarily resulted from unrecognizable text, invalid syntax, or ambiguous operator patterns, which together accounted for the largest share of failures. Percentages are calculated both by record count and by the number of distinct source units.

3.2. Performance of the unit harmonization function

Out of 157,717,962 standardized records encompassing 382 distinct UCUM units, the harmonization function successfully harmonized 83.3 % of results in less than 2 min, leading to 39 and 42 distinct reference units under the Conventional and SI systems, respectively (Table 2). Within these harmonized records, nearly half (≈ 50.5 %, ~ 80 million) required no numeric conversion, either because the source unit was equivalent to the reference unit with the same dimension and magnitude (7.0–7.8 %) or was already identical to the harmonized unit (42.6–43.6 %).

Summary of harmonization outcomes for 157.7 million laboratory result records, presented separately for SI and conventional unit systems. Overall, 83.4 % of records were harmonized to a reference unit. Among these, approximately half (50.4–50.6 %) required no conversion because the source unit was equivalent or already harmonized, while the remaining 32.8–33 % underwent numeric conversion. Regular conversions accounted for the majority of converted records, with mass-molar conversions representing 1.9 % of conventional and 13.5 % of SI-based results. Unharmonized records (16.6 %) were primarily due to mismatched dimensions or incompatible arbitrary units. Percentages reflect proportions within each unit system.

Conversely, approximately one-third of the original records (~ 33 %, ≈ 52 million) required unit conversion. The majority of these conversions were regular conversions (31.1 % under Conventional, 19.3 %

Table 2

Results of the Harmonization Function Across SI and Conventional Unit Systems.

Unit System	Conventional		SI	
	#	%	#	%
Harmonized	131,465,447	83.4 %	131,465,471	83.4 %
Not converted	79,488,983	50.4 %	79,721,116	50.6 %
equivalent source unit	12,271,914	7.8 %	10,995,783	7.0 %
harmonized source unit	67,217,069	42.6 %	68,725,333	43.6 %
Converted	51,976,464	33 %	51,744,355	32.8 %
mass-molar conversion	2,921,251	1.9 %	21,251,533	13.5 %
regular conversion	49,055,213	31.1 %	30,492,822	19.3 %
Not harmonized	26,227,241	16.6 %	26,227,217	16.6 %
Different dimensions	22,842,968	14.5 %	22,841,237	14.5 %
Different arbitrary units	2,923,187	1.9 %	2,924,894	1.9 %
No reference unit available	331,185	0.2 %	331,185	0.2 %
No molecular weight available	92,803	0.1 %	92,803	0.1 %
Invalid-UCUM units	37,086	0.0 %	37,086	0.0 %
Special conversion required	12	0.0 %	12	0.0 %
Grand Total	157,692,688			

under SI). Mass-molar conversions were substantially more frequent in the SI system (13.5 %) than in the Conventional system (1.9 %), reflecting the greater alignment of SI units with molar-based measurements and the higher prevalence of conventional units in real-world laboratory data.

A total of 16.6 % of records (≈ 26 million) could not be harmonized under either system. The primary reason for non-harmonization was dimensional mismatch (14.5 %), where the source and reference units represented different measurement properties. A notable example is the conversion from “%” as a source unit to $10^9/L$ as the reference unit for absolute (rather than relative) differential White Blood Cell (WBC) counts, accounting for 9.8 million records. These cases largely stem from inconsistencies between assigned LOINC codes and source units in the original data.

Similar to the validation of the standardization function, we assessed the correctness of numeric conversions performed by the harmonization function by comparing its outputs to the official UCUM conversion API, which provides reference calculations for UCUM-compliant units. Numeric equivalence was evaluated using a tolerance criterion of $|\Delta| < 1 \times 10^{-5}$, corresponding to five-decimal reporting precision. Under this criterion, all converted values matched their UCUM-API references, achieving 100 % concordance and confirming the accuracy of magnitude-based rescaling operations. These results not only validate the reliability of our conversion algorithm but also demonstrate the robustness of the integrated unit parsing function, which accurately derives essential metadata—such as dimension, magnitude, and other unit parameters—for previously unseen units when these attributes are not readily available in the included pre-parsed dataset.

3.3. Impact on LOINC code

During the harmonization process, mass-molar LOINC-unit mismatches were identified and corrected in 2.8 million records (1.8 % of harmonized results). For all cases where a mass-molar conversion was applied, the corresponding LOINC code was updated to reflect the new property type.

These updates resulted in measurable changes to the distribution of LOINC codes in the harmonized dataset compared to the original input. Additionally, as part of the harmonization workflow, individual LOINC codes were aggregated into clinically meaningful LOINC groups, reducing code fragmentation and enabling feature consolidation for downstream analytics. Fig. 3 visualizes these changes, illustrating how initial LOINC codes were removed, updated, or aggregated through harmonization. The diagram highlights both reduced code

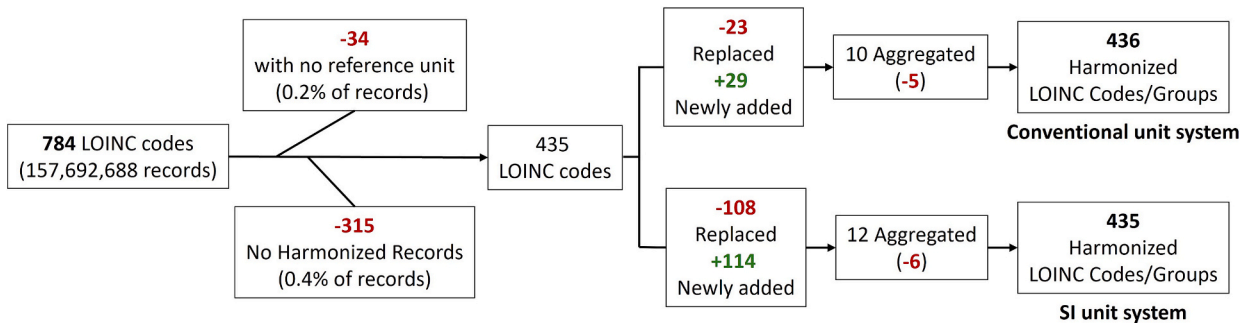


Fig. 3. Impact of harmonization on LOINC code distribution and grouping. This figure illustrates changes in the number of LOINC codes from the original dataset to the harmonized output, including adjustments due to mass–molar conversions and aggregation into LOINC groups. Negative values (in red) indicate reductions resulting from consolidation or code replacement, while positive values (in green) represent additions from newly assigned LOINC codes after property changes. Overall, harmonization reduced code fragmentation and aligned laboratory results to standardized LOINC group structures, improving semantic interoperability and supporting feature aggregation for downstream analytics. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

fragmentation and the semantic alignment achieved through systematic harmonization.

3.4. Impact on result value distribution

The structural adjustments at the LOINC code level set the stage for evaluating how harmonization affected the distribution of laboratory results across the updated LOINC codes and groups by three metrics of distributional quality. For interpretation, we classified distributional changes as Improved, Worsened, or Steady using practical thresholds. For IQR and outlier proportion, a relative change of $\geq 5\%$ was considered meaningful, whereas for mode count, any absolute change in the number of peaks indicated a structural shift in the distribution. Thus, an ‘Improved’ distribution reflects narrower spread, fewer outliers, or more coherent modality; a ‘Worsened’ distribution reflects the opposite.

Fig. 4 summarizes the results for common LOINC groups ($n \geq 10,000$ observations), analyzed separately for SI and conventional unit systems. For both systems, the majority of groups exhibited steady distributional characteristics, while 11–23 % of groups showed improvement across one or more metrics, and only 2–12 % displayed worsening. Among all

metrics, mode count showed the clearest benefit from harmonization, with reductions in artificial multimodality for most analytes; only three groups demonstrated increased complexity.

Fig. 5 illustrates representative examples of laboratory value distributions before and after harmonization, highlighting cases of improvement, stability, and occasional worsening across different analytes (e.g., white blood cell counts, thyrotropin, and creatinine) under both SI and Conventional systems.

4. Discussion

4.1. Principal findings

This study introduces and evaluates two scalable, open-source R functions designed to standardize and harmonize laboratory unit representations across heterogeneous, real-world, multi-source EMR datasets. The approach leverages the machine-readability of UCUM for unit standardization and the aggregability of LOINC groups for semantic harmonization.

Applied to over 160 million records from the Intego-II database, the

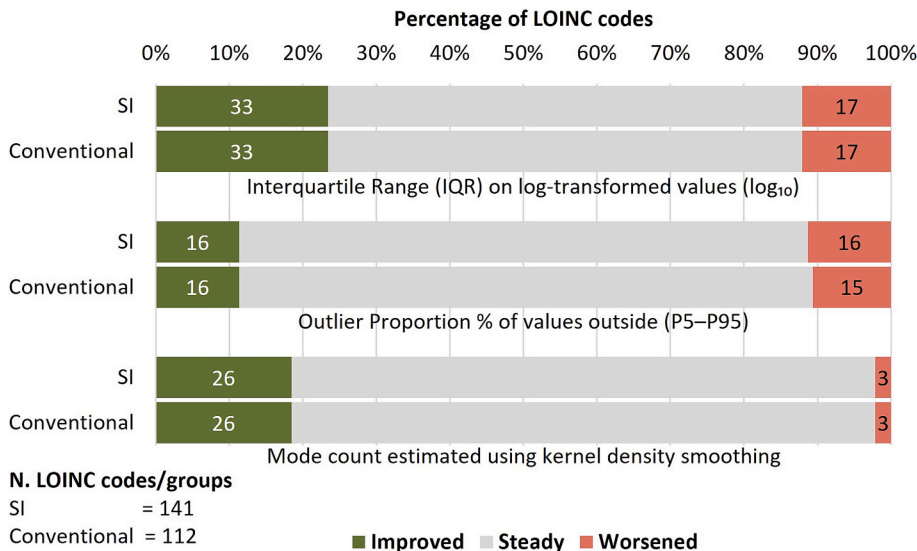


Fig. 4. Changes in Distributional Metrics After Harmonization by Unit System The figure shows the impact of harmonization on laboratory value distributions across LOINC groups with $\geq 10,000$ observations, stratified by unit system (SI vs conventional). Metrics included log-transformed IQR, outlier proportion (values outside the 5th–95th percentile), and mode count (number of distributional peaks). Categories represent the direction of change post-harmonization: Improved (green), Steady (gray), or Worsened (red), based on practical thresholds ($\geq 5\%$ for IQR and outlier proportion, and any change for mode count). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

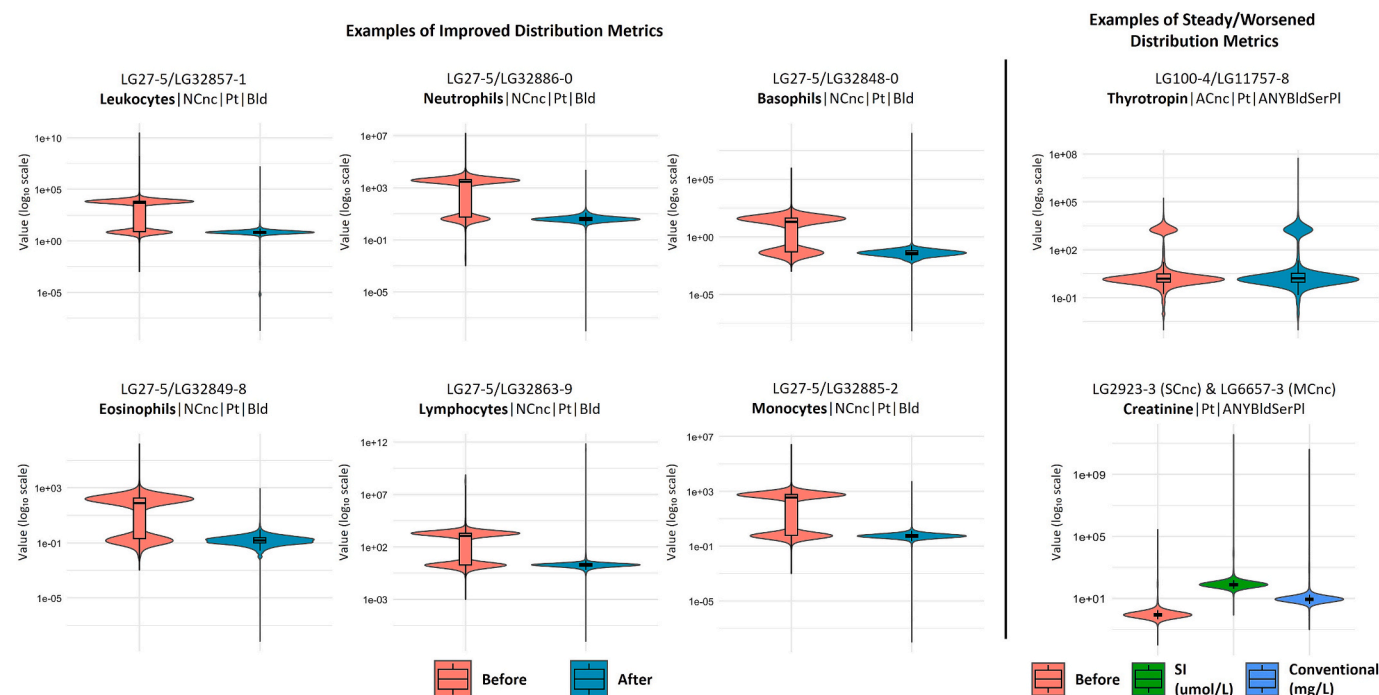


Fig. 5. Distribution of Laboratory Values Before and After Harmonization Across LOINC Groups (Log₁₀ Scale). This figure illustrates the distribution of laboratory test results before and after unit harmonization for multiple common LOINC code groups. Each panel corresponds to a specific LOINC group. The horizontal axis represents the harmonization stage, and the vertical axis displays laboratory values on a log₁₀ scale to accommodate the wide range of measurements. Violin plots depict the full distribution of values for each stage, with embedded boxplots indicating the median and interquartile range (IQR). The left panels show violin plots for six total and differential White Blood Cell (WBC) count tests, all of which demonstrate improvement across all distribution metrics after harmonization. The right panels include tests where changes were minimal or, in some cases, unfavorable. For example, Thyrotropin exhibits a wider IQR post-harmonization while maintaining similar mode count and outlier proportion. Creatinine serves as a unique example, illustrating comparisons between two unit harmonization systems—SI and conventional (MCnc or SCnc property groups). Although the two systems differ in scale, creatinine values remain largely consistent across all distribution metrics.

approach achieved 96.2 % standardization and 83.4 % harmonization, with outputs validated through official services. Processing was completed within minutes, confirming efficiency. These results demonstrate the feasibility and importance of automated unit cleaning as a critical foundation for reliable analytics and machine learning pipelines.

4.2. Comparison with prior work

Previous research addressing laboratory unit variability has typically focused on one-off mapping projects or institution-specific initiatives that targeted only one aspect—either standardization or harmonization—often without scalable automation. In the context of standardization, one study reported increasing UCUM compliance from 40 % to 96 % in a large multi-source dataset [12]. However, this solution was implemented locally on a private database and was not released as open-source software, limiting reproducibility in other settings. Other approaches have framed unit standardization as a terminology-matching problem, applying fuzzy search techniques or, more recently, language models combined with retrieval algorithms and manual expert review [18,19]. While these methods show potential for semantic alignment, they do not exploit the compositional structure of UCUM units, which is essential for reliable numeric conversions. By contrast, our approach interprets units as computational expressions derived from the seven UCUM base units, applying deterministic mathematical rules for accurate, machine-readable standardization rather than relying on approximate text-based methods.

Harmonization efforts have followed a different trajectory, typically focusing on converting pre-standardized units to common reference units within large-scale data repositories. Examples include initiatives in the Veterans Health Administration and the National COVID Cohort

Collaborative datasets, which relied heavily on curated canonical unit lists and static conversion tables [3,20]. Although these approaches improved consistency in specific contexts, they lacked formalized algorithms that leverage the machine-readable nature of UCUM and often depended on pre-mapped OMOP concepts or even non-UCUM units such as µg/dL. This overlooks the critical prerequisite that real-world units must first be standardized, ultimately limiting automation and scalability. A further challenge in prior harmonization strategies is the absence of objective criteria for selecting reference units. Most relied on subjective expert consensus, introducing variability and inconsistency. Tools such as LUMA attempted to mitigate this by offering multiple unit options per LOINC code based on property matching. While helpful for manual workflows, such approaches leave ambiguity unresolved [21].

In contrast, our approach integrates both standardization and harmonization in a unified framework. It converts heterogeneous real-world units into UCUM-valid expressions and, based on user preference for SI or Conventional systems, harmonizes results to systematically chosen reference units following IFCC-IUPAC recommendations and OMOP standards. By leveraging UCUM's computational structure, the process applies deterministic, rule-based algorithms validated through official services, ensuring precision and reproducibility. Both functions are released as an open-source package to support scalability and reuse.

4.3. Interpretation and Implications

This work demonstrates that large-scale standardization and harmonization of laboratory units can be achieved accurately and efficiently through fully automated processes suitable for integration into routine Extract-Transform-Load (ETL) workflows. Harmonization improved unit consistency and reduced LOINC code fragmentation,

enabling robust feature construction for analytics and machine learning while minimizing bias and numeric errors to support reliable, generalizable models. The process also served as a quality-control checkpoint by detecting dimensional mismatches in 14.5 % of unharmonized records and correcting mass-molar LOINC inconsistencies in 1.8 % of harmonized records. Because of the deterministic nature of the functions, expert annotation would be most valuable for reviewing the relatively small subset of records flagged as failures, rather than revalidating the entire standardized or harmonized dataset.

4.4. Limitations and future directions

While this work demonstrates strong performance and scalability, several limitations remain. First, approximately 3.8 % of records could not be standardized, mainly due to malformed or unrecognizable unit strings. Some of these reflected language-specific terms, such as Dutch words like “miljoen” for “million,” whereas our function relied on international English-based resources, limiting performance on data containing local-language unit terms. Extending the approach with multilingual dictionaries or user-supplied synonym tables will improve generalizability and is planned for future iterations. Second, although the standardization function achieved near-perfect correctness, a small proportion of invalid units (0.03 %) passed validation checks. These units were, however, excluded from harmonization due to incompatibility with conversion algorithms.

The harmonization function also relies on accurate LOINC mappings and consistency between units and reported values. In rare cases where mismatches occurred—such as reporting results in [IU]/L while specifying m[IU]/L—conversions led to scaling distortions, as observed in the Thyrotropin distribution (Fig. 2). These artifacts highlight the need for integrated quality-control steps to assess plausibility after harmonization. Incorporating automated plausibility checks as a post-processing layer would help mitigate such issues [22].

Finally, this study focused on two key aspects of unit cleaning—standardization and harmonization—but these are only part of the broader challenge of laboratory data quality. Recognizing this, both functions are integrated into the lab2clean R package, which also includes functions for standardizing and validating result values. Expanding this automated cleaning framework with covering additional variables and quality dimensions represents an important direction for future research.

Ethics statement

The Intego-II database procedures were approved by the KU Leuven Ethics Committee (approval number S68557) and the Belgian Information Security Committee, specifically the Social Security and Health Chamber in their decision number 13.026, dated March 19th, 2013, with the most recent update in decision number 23.424, dated December 5th, 2023.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT as a writing assistant to improve the language and clarity of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Funding sources

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

CRedit authorship contribution statement

Ahmed Medhat Zayed: Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Ilias Sarikakis:** Writing – review & editing, Validation, Software, Methodology, Data curation. **Nicolas Delvaux:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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